

Bi-Q

A simple two-knob hi-fi tone control for single tracks.



Overview

Bi-Q is a bass and treble tone control in the spirit of the classic hi-fi tone circuit. Its curves are broad 6 dB per octave shelves that keep rising gently towards the extremes rather than flattening off, so a little goes a long way and the middle of the spectrum is left alone.

Each control has a TURNOVER switch that sets where it is half-way to its full effect. With the bass at 75 Hz and the treble at 6 kHz it only touches the very ends; at 300 Hz and 1.5 kHz it reshapes much more of the sound.

A slow drift makes the turnover points and gains wander very slightly, left and right independently, like real components warming up.

Reach for it for

- Quick, musical bass and treble shaping on a single track
- Adding weight or air without touching the mids
- Gentle broad tilts on vocals, keys and acoustic instruments
- Taming boomy or harsh recordings with a broad cut

Controls

Tone

BASS	Boosts or cuts the low end, -12 to +12 dB in 0.5 dB steps (hold Shift for free movement). The curve is a broad shelf, so the lowest notes move most.
TURNOVER (bass)	Where the bass control is half-way to its full effect: 75, 150 or 300 Hz. Click a label to pick it, or click / drag / scroll the switch. 75 Hz for just the bottom; 300 Hz to reshape the whole low end.
TREBLE	Boosts or cuts the top end, -12 to +12 dB in 0.5 dB steps (Shift for free movement).
TURNOVER (treble)	Where the treble control is half-way to its full effect: 1.5k, 3k or 6k. 6k adds or removes air; 1.5k brightens or darkens much more of the sound.
IN	Switches the tone controls in or out (in by default). When out, the tone section is dimmed and the audio passes flat; OUTPUT still works.

Output

OUTPUT	Output level, -12 to +12 dB. Use it to match the level after a boost or cut.
LEVEL	A row of ten LEDs showing the output peak level, from -30 dB up to just under 0 dBFS (green, then amber, then red).

Drift

The bass and treble turnover points and gains wander very slightly and slowly, separately on the left and right sides.

DRIFT	How much drift, 0 to 100% (50% by default). 50% is the standard amount, 100% is twice that and 0 switches it off. The wander also speeds up a little as you turn it up.
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IN / OUT

TRIM	Input level before anything else, from -24 to +12 dB. A single click on the knob (without dragging) switches GUARD on or off; its LED sits at the knob's upper right.
GUARD	On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. The LED is red while it stands by and glows brighter while it is working.
SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper at -0.3 dBFS (no latency), or a lookahead limiter holding the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). The CLIP key flashes red while the clipper is working.

The header

S / M / L	Panel size: small, medium or large.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Tips

- Start with small moves: because the shelves keep rising towards the extremes, 2 or 3 dB is often all a track needs.
- For air without harshness, set the treble TURNOVER to 6k and add a few dB; for a darker, warmer tone, cut with it at 1.5k.
- Cutting BASS with TURNOVER at 300 Hz is a quick way to thin out a muddy guitar or piano in a busy mix.
- Switch IN on and off at matched levels (use OUTPUT) to judge whether the change really helps.

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